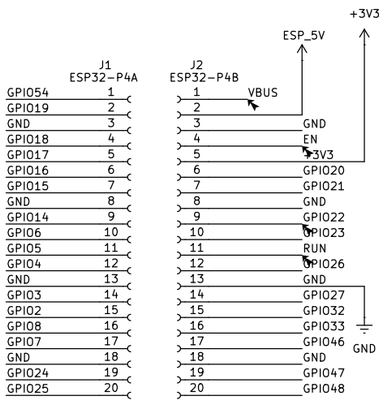
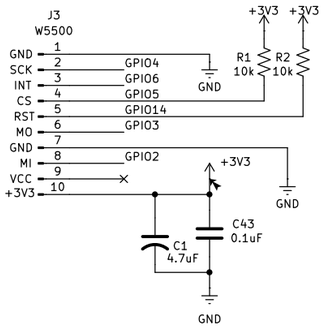
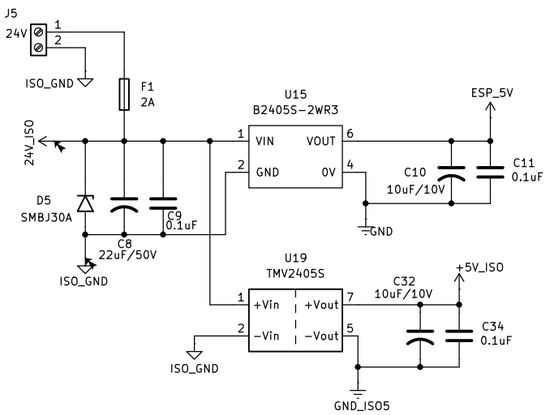
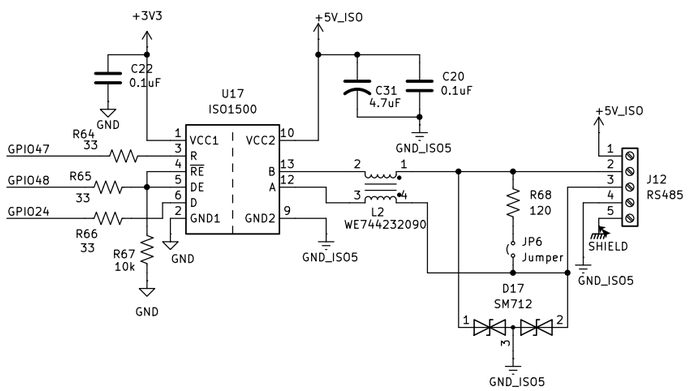
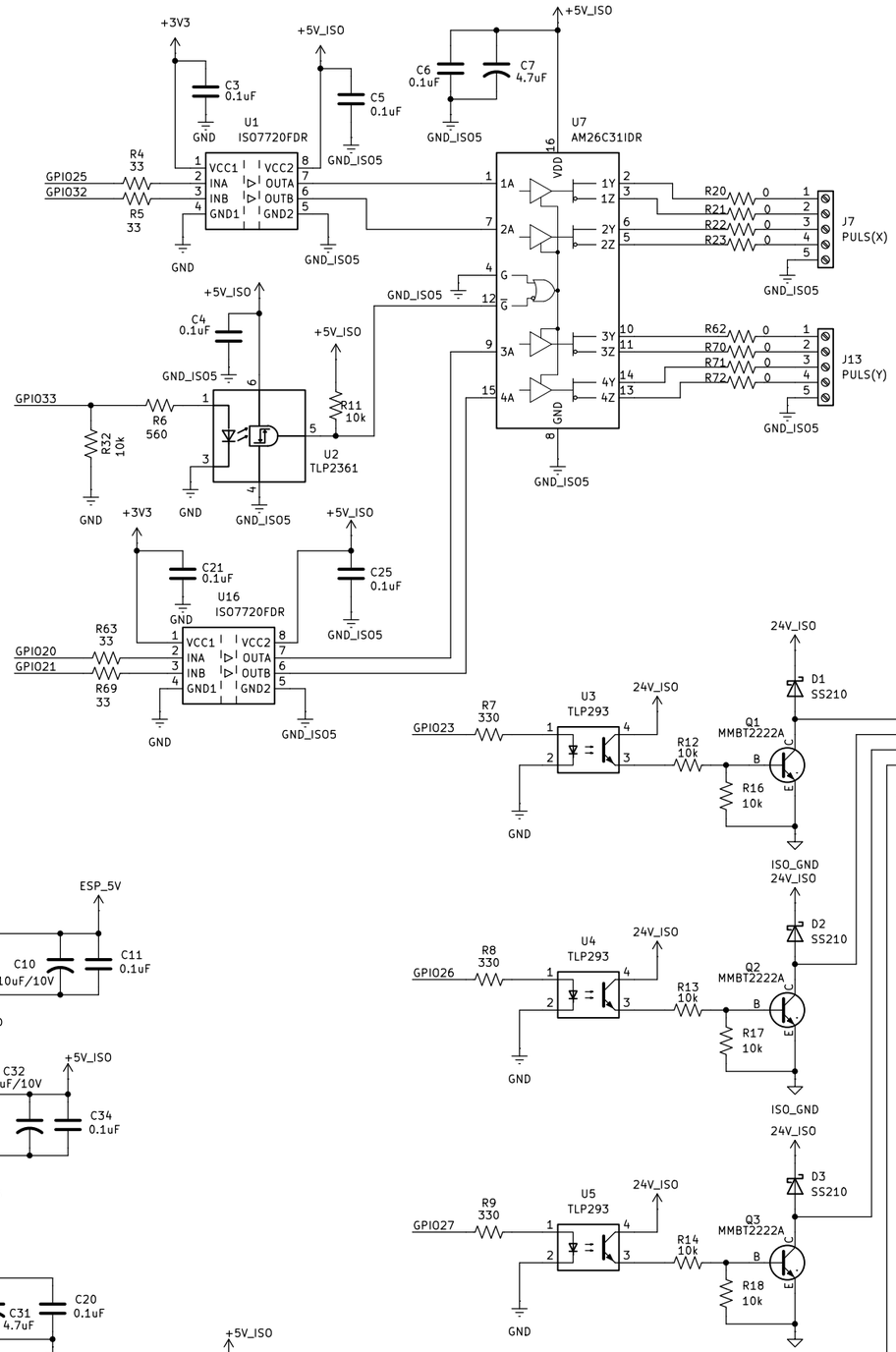
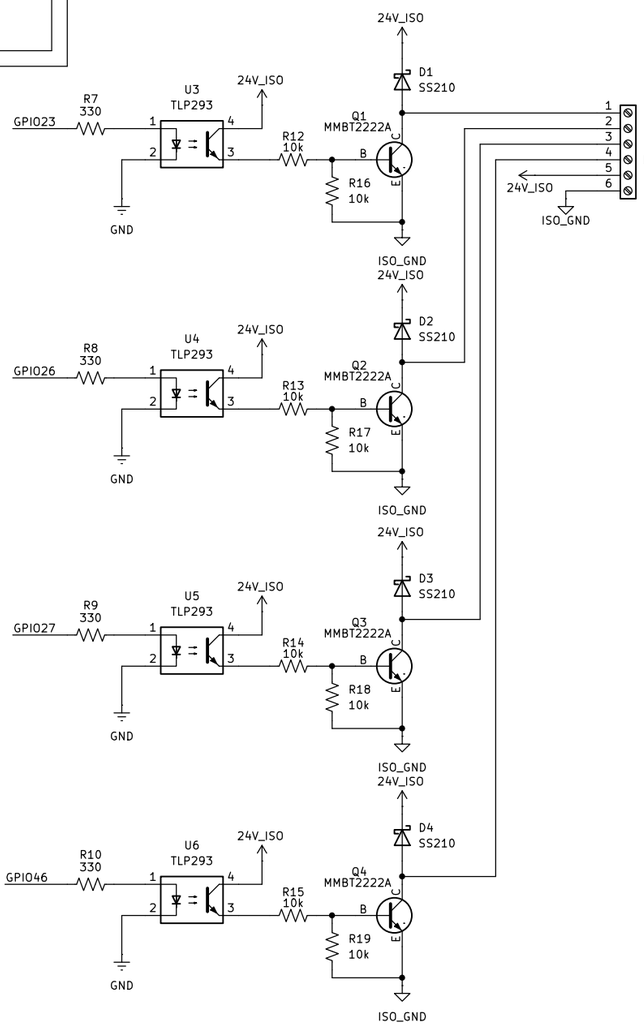

J1
ESP32-P4A
J2
ESP32-P4B
ESP_5V
+3V3
VBUS
GND
EN
RUN
GPIO54
GPIO19
GPIO18
GPIO17
GPIO16
GPIO15
GPIO14
GPIO6
GPIO5
GPIO4
GPIO3
GPIO2
GPIO8
GPIO7
GPIO24
GPIO25
GPIO20
GPIO21
GPIO22
GPIO23
GPIO26
GPIO27
GPIO32
GPIO33
GPIO46
GPIO47
GPIO48

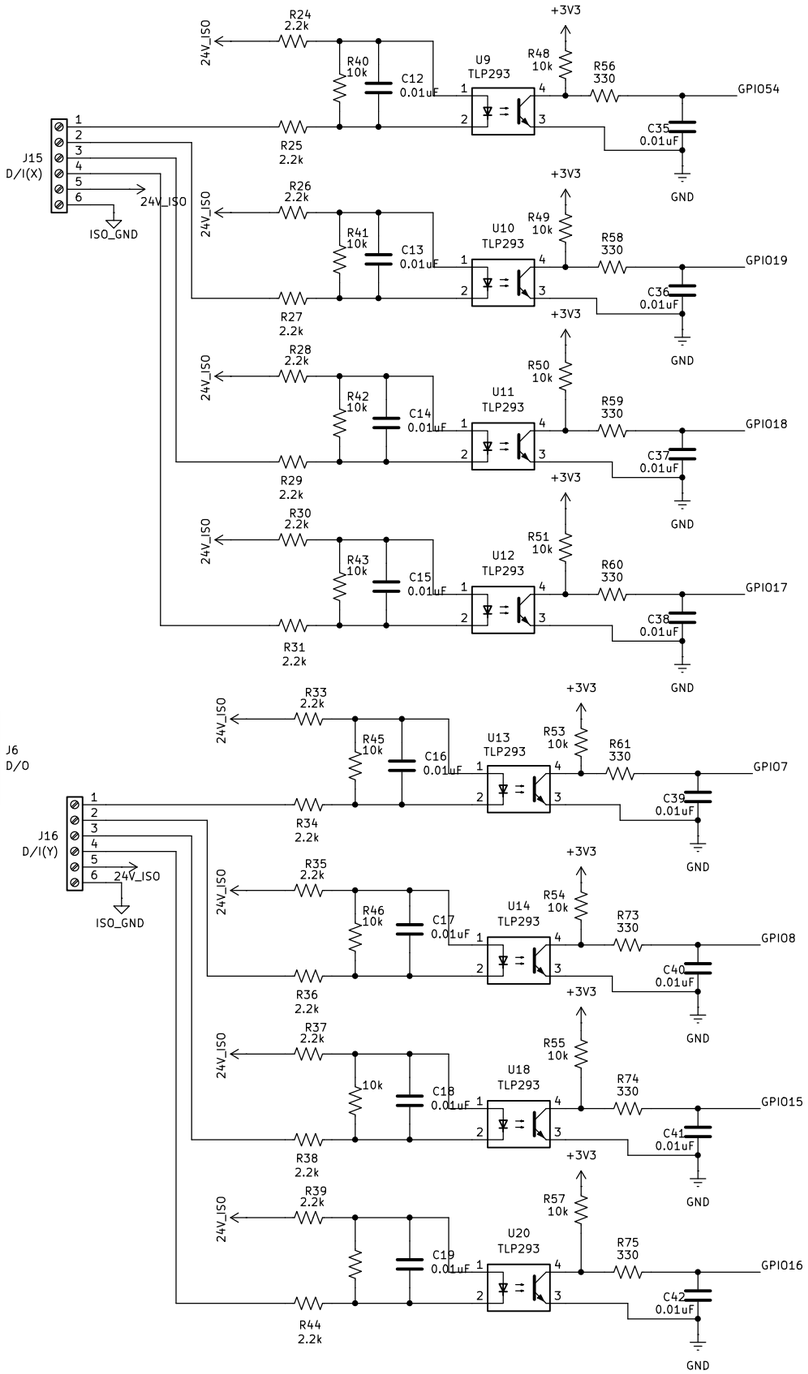

J3
W5500
GND
SCK
INT
CS
RST
MO
MI
VCC
+3V3
R1 10k
R2 10k
C1 4.7uF
C43 0.1uF


J5
24V
ISO_GND
F1
2A
24V_ISO
D5
SMBJ30A
C8
22uF/50V
C9
0.1uF
U15
B2405S-2WR3
VIN
VOUT
GND
0V
ESP_5V
C10
10uF/10V
C11
0.1uF
U19
TMV2405S
+Vin
-Vin
+Vout
-Vout
C32
10uF/10V
+5V_ISO
C34
0.1uF
GND_ISO5


+3V3
+5V_ISO
C22
0.1uF
U17
ISO1500
C31
4.7uF
C20
0.1uF
GPIO47
GPIO48
GPIO24
R64
33
R65
33
R66
33
R67
10k
VCC1
VCC2
R
RE
DE
D
GND1
GND2
B
A
L2
WE744232090
R68
120
JP6
Jumper
SHIELD
D17
SM712
J12
RS485
GND_ISO5


+3V3
+5V_ISO
C3
0.1uF
C5
0.1uF
C6
0.1uF
C7
4.7uF
U1
ISO7720FDR
U7
AM26C31IDR
GPIO25
GPIO32
R4
33
R5
33
VCC1
INA
INB
GND1
VCC2
OUTA
OUTB
GND2
VDD
1A
2A
3A
4A
1Y
1Z
2Y
2Z
3Y
3Z
4Y
4Z
G
R20
R21
R22
R23
J7
PULS(X)
R62
R70
R71
R72
J13
PULS(Y)
C4
0.1uF
R6
560
R32
10k
R11
10k
U2
TLP2361
GPIO33
C21
0.1uF
C25
0.1uF
U16
ISO7720FDR
GPIO20
GPIO21
R63
33
R69
33
GND_ISO5


24V_ISO
D1
SS210
D2
SS210
D3
SS210
D4
SS210
U3
TLP293
U4
TLP293
U5
TLP293
U6
TLP293
GPIO23
GPIO26
GPIO27
GPIO46
R7
330
R8
330
R9
330
R10
330
R12
10k
R13
10k
R14
10k
R15
10k
Q1
MMBT2222A
Q2
MMBT2222A
Q3
MMBT2222A
Q4
MMBT2222A
R16
10k
R17
10k
R18
10k
R19
10k
J6
D/O
ISO_GND


J15
D/I(X)
J16
D/I(Y)
24V_ISO
ISO_GND
R24
2.2k
R25
2.2k
R26
2.2k
R27
2.2k
R28
2.2k
R29
2.2k
R30
2.2k
R31
2.2k
R33
2.2k
R34
2.2k
R35
2.2k
R36
2.2k
R37
2.2k
R38
2.2k
R39
2.2k
R44
2.2k
R40
10k
R41
10k
R42
10k
R43
10k
R45
10k
R46
10k
R48
10k
R49
10k
R50
10k
R51
10k
R53
10k
R54
10k
R55
10k
R57
10k
U9
TLP293
U10
TLP293
U11
TLP293
U12
TLP293
U13
TLP293
U14
TLP293
U18
TLP293
U20
TLP293
C12
0.01uF
C13
0.01uF
C14
0.01uF
C15
0.01uF
C16
0.01uF
C17
0.01uF
C18
0.01uF
C19
0.01uF
R56
330
R58
330
R59
330
R60
330
R61
330
R73
330
R74
330
R75
330
C35
0.01uF
C36
0.01uF
C37
0.01uF
C38
0.01uF
C39
0.01uF
C40
0.01uF
C41
0.01uF
C42
0.01uF
GPIO54
GPIO19
GPIO18
GPIO17
GPIO7
GPIO8
GPIO15
GPIO16